

The Effect of Music on Concentration in Healthy Young Adults

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Abstract

This study examined whether listening to music affects concentration in healthy young adults (ages 18–35). Islander volunteers (N = 73 analyzed, after 2 withdrawals) were block-randomized by village to either a music condition (10 minutes of their favorite-genre music) or a no-music control, with attention measured by a timed attention test immediately before and after the treatment period; the outcome was the difference in attention score. The registered primary estimate showed the music group's attention score had a 1.17 point greater mean change than the control group's (95% CI: −1.28 to 3.62 points), a difference that was not statistically significant ($p = 0.344$). We conclude that this experiment found no reliable evidence that listening to music affects short-term attention performance relative to sitting without music.

Introduction

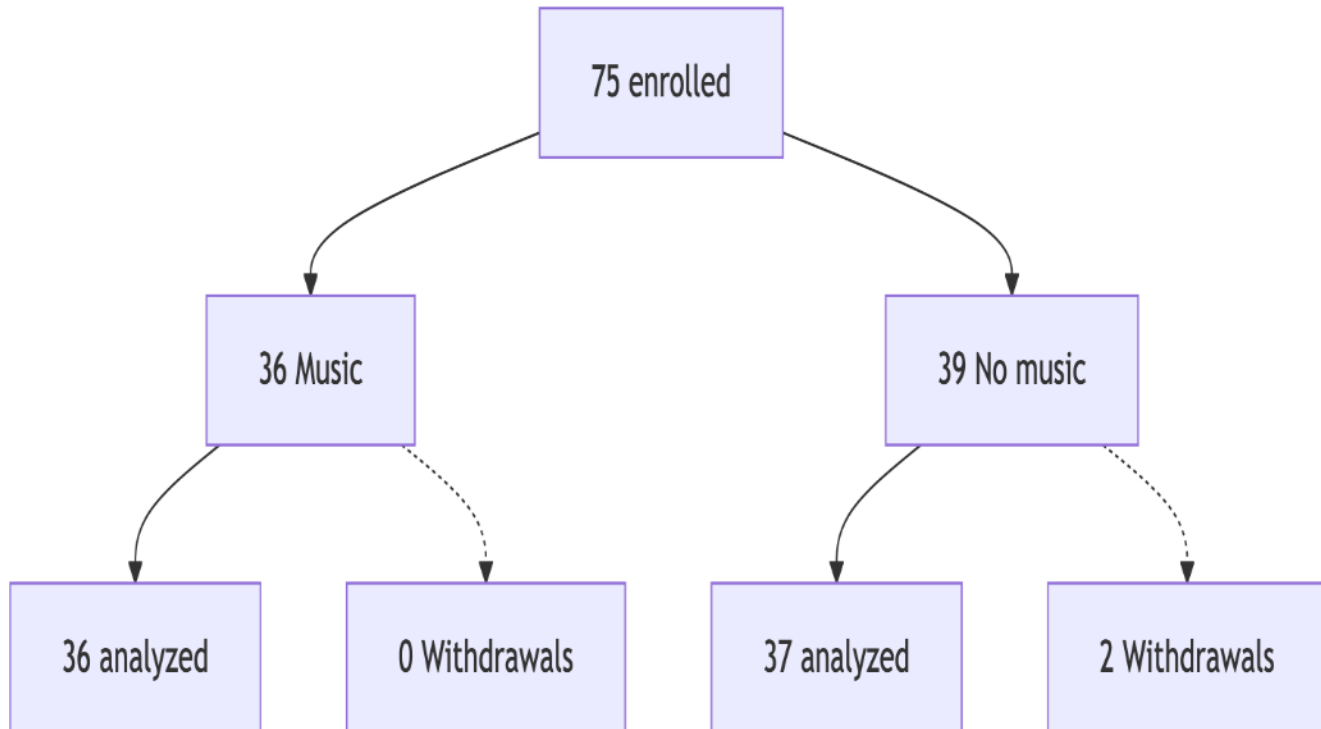
Many people, especially students, listen to music while completing work or other tasks that require concentration, but it isn't clear whether music improves concentration, interferes with it, or has little to no effect. Understanding this relationship could help young adults make a more informed decision about whether to listen to music while doing tasks that require sustained attention. Does listening to music affect concentration in healthy young adults? In particular, we are interested in whether listening to music leads to a change in performance on an attention test compared to taking the attention test without music beforehand.

Methods: participants, design, and protocol

Recruitment was done via newspaper among the islands. The eligibility criteria was determined by age range, selecting for Islanders between the ages of 18-35 years old. Consent was received before beginning the trials and Islanders were allowed to withdraw consent during trials; data from participants who withdrew were excluded from analysis. The population to which the results reasonably may apply are young adults from 18-35 years old, since that is the population we tested for. The experimental and response unit are both the participant. The treatment structure is one treatment factor with two levels. The factor is the listening condition, with one treatment being the Islander listening to their favorite music genre for 10 min and the control being to sit without music for 10 minutes. Attention is measured immediately before and after the treatment period. The allocation structure used in this study was one factor block randomization. For this study, participants were blocked by villages on the same island. Within each village, participants were randomly assigned where half received music treatment and the other half received no treatment. The blocking factor used in this study was the villages. Participants from the same villages were grouped together before treatment assignment, and random assignment between music or no music conditions were assigned within each village. Blocking by villages helped control for potential differences among villages, such as cultural differences and music preferences, which may affect their results in the attention tests. Each Islander first completed a 10-minute attention test to obtain a baseline score. Participants assigned to the treatment condition then listened to their favorite genre of music for 10 minutes, while participants assigned to the control group sat without music for 10 minutes. Immediately after, all participants completed a second 10-minute attention test. The response variable was the change in attention score from baseline to the post-treatment measurement. No blinding was used. We used a significance level of $\alpha = 0.05$ and aimed for a target power of 0.80. For precision, we targeted a 95% confidence interval with an expected width of approximately 7.76 points. Based on our pilot study, which produced an estimated standard deviation of 6.84 points, we used a more conservative plausible standard deviation of 8.25 points for our planning calculations.

Data and Results

Enrolled	Measured	Excluded	Analyzed
75	75	2	73



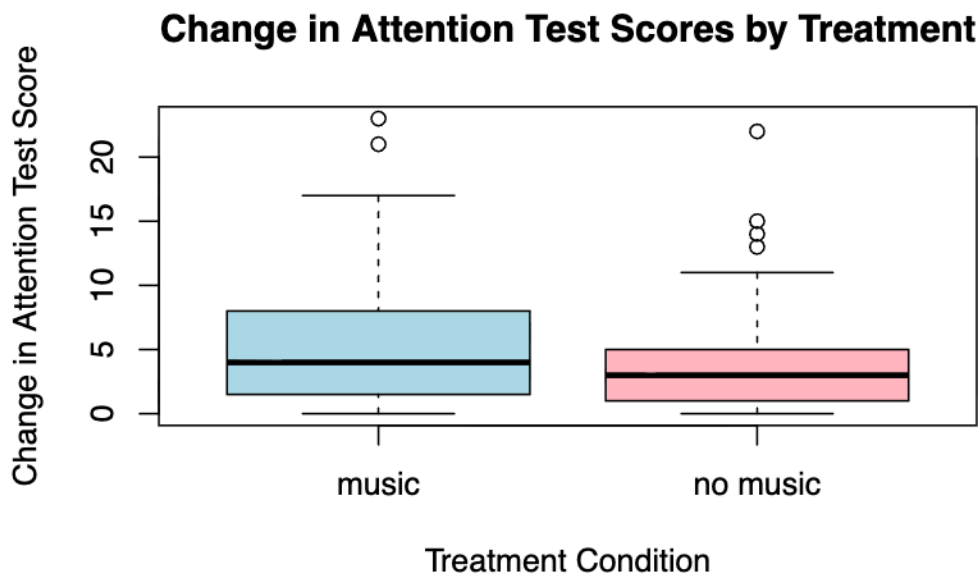
A total of 75 participants were enrolled and assigned to treatment. Two participants withdrew before producing a usable primary response, leaving 73 participants in the final analysis. The final analyzed sample consisted of 36 participants in the music condition and 37 participants in the no-music condition. The mean change in attention score was 5.61 points in the music group and 4.32 points in the no-music group. The corresponding standard deviations were 5.69 and 4.97 points, respectively.

Village	No Music	Music
Akkeshi	4	4
Arcadia	3	4
Biruwa	4	4
Hayarano	4	4
Kiyobico	4	4
Nelson	4	4
Reading	4	4
Shinobi	4	4
Takazaki	6	4

The only block in our treatment groups were the villages. Our two treatment groups were music and no music for control. Although most village blocks were approximately balanced between treatment groups, there were some missing outcomes and other data collection errors. Akkeshi contained one fewer usable

observation because of a withdrawal that could not be fully replaced via the backup pool. Takazaki contained additional observations because more participants than planned were assigned and measured during data collection. These observations were retained according to the frozen-data protocol.

```
boxplot(
  difference_in_scores ~ treatment,
  data = island_usable,
  xlab = "Treatment Condition",
  ylab = "Change in Attention Test Score",
  main = "Change in Attention Test Scores by Treatment",
  col = c("lightblue", "lightpink")
)
```



Treatment	Participants	Mean	SD	Median	Min	Max
Music	36	5.61	5.69	4	0	23
No Music	37	4.32	4.97	3	0	22

Analysis

Effect of music on change in attention score

Outcome	Music	No music	Adjusted treatment effect	95% CI	P value
Change in attention score, mean $\pm$ SD (n)	5.61 $\pm$ 5.69 (36)	4.32 $\pm$ 4.97 (37)	1.17 points	-1.28 to 3.62	0.344

Adjusted treatment effect estimated from the linear model:

$$\text{difference in scores} \sim \text{village} + \text{treatment}$$

P value is for the between-treatment comparison after adjusting for village.

The hypothesis test evaluated

$$H_0 : \beta_{\text{music}} = 0$$

against

$$H_A : \beta_{\text{music}} \neq 0$$

The change in Attention Score was analyzed using a Linear Model where village was included as a blocking variable and treatment compared the music condition with the no-music control condition.

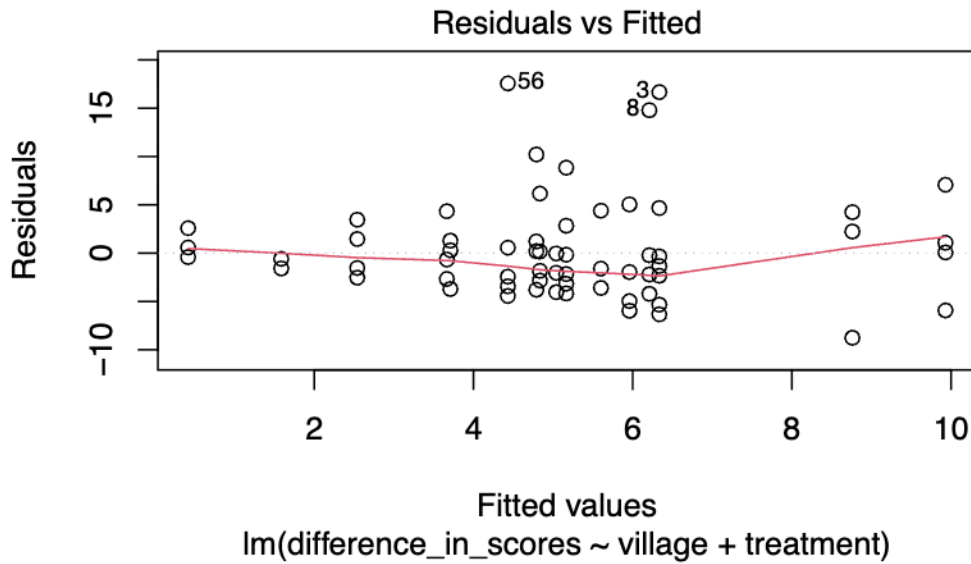
Overall effects of village and treatment

Effect	df	F statistic	P value
Village	8, 63	1.43	0.203
Treatment	1, 63	0.91	0.344

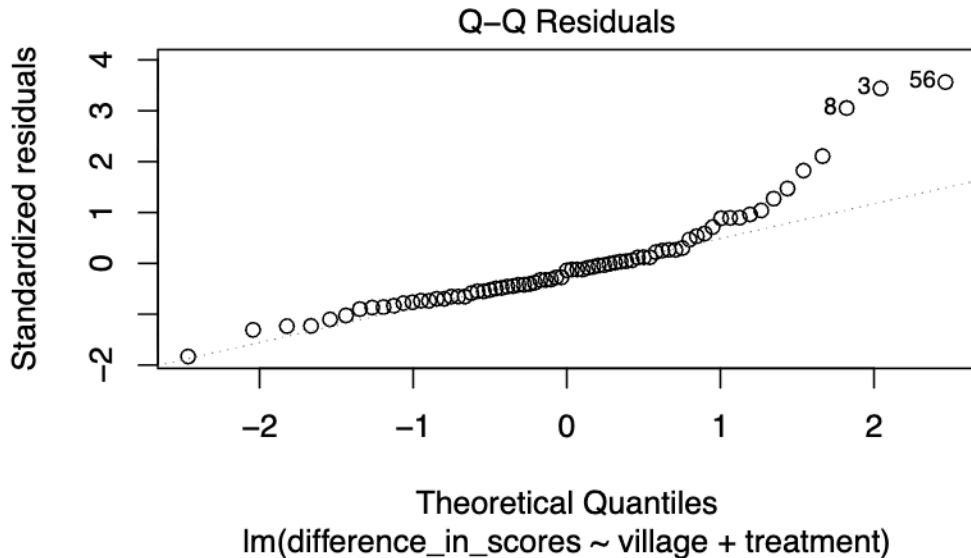
After adjusting for village, the music group had a 1.17-point greater mean change in attention score than the no-music group (95% CI: -1.28 to 3.62). This difference was not statistically significant, $F(1, 63) = 0.91$, $p = 0.344$. The overall village block effect was also not statistically significant, $F(8, 63) = 1.43$, $p = 0.203$.

The primary treatment contrast compared the mean absolute change in attention test score between the music and no-music conditions. Factorial interactions were not applicable because the experiment included only one treatment factor with two levels. Village was included as a blocking factor to account for possible systematic differences among villages. After accounting for village, the treatment effect was not statistically significant.

```
plot(music_model, which = 1) # Residuals vs Fitted
```



```
plot(music_model, which = 2) # Normal Q-Q
```



The assumptions of the linear model were evaluated using the study design and residual diagnostics. Independence was supported by the randomized assignment of individual participants within village blocks, with each participant contributing one primary response. A residuals-versus-fitted plot was examined to assess constant variance and model form. The residuals appeared to be reasonably scattered around zero without a strong systematic pattern, suggesting that the constant-variance assumption was reasonable. A normal Q-Q plot was used to assess the normality of the residuals. The residuals were approximately consistent

with a normal distribution, although some deviation in the tails was present. No severe departures from normality were observed. Overall, the diagnostic checks did not indicate major violations of the linear-model assumptions, so the fitted model was considered appropriate for the primary analysis.

Cook's Distance (N = 73):

$$4/73 \approx 0.0548$$

Potentially Influential Observations: 8, 56, and 3; Their Cook's distances were 0.1907, 0.1502, and 0.1552, respectively, all exceeding the screening threshold of 0.0548. These observations were examined but were not removed because there was no evidence that they resulted from data-entry errors or protocol violations.

Conclusion

The study examined whether listening to favorite music for 10 minutes affects concentration in young adults compared with sitting without music. The results did not provide sufficient evidence of a statistically significant effect of music on short-term attention performance. Although the music group exhibited a slightly greater mean change in attention score than the no-music group, the observed difference was small and not statistically significant. Therefore, the findings do not support the conclusion that listening to favorite music for 10 minutes meaningfully affects short-term attention performance in young adults.

Discussion

There are several limitations to consider when interpreting these results. First, participants were volunteers rather than a random sample of the broader population, which limits the generalizability of the findings. Second, the treatment period lasted only 10 minutes, and attention was measured immediately afterward. Therefore, the results apply only to short-term attention performance and do not indicate whether any effect of music would persist over longer periods of time. In addition, participants in the music condition listened to music from their preferred genre, so the specific music exposure varied across individuals. Because treatment assignment was randomized within village blocks, causal conclusions can be made about the effect of the assigned music condition for participants in this experiment. However, because the sample consisted of young adult Islander volunteers ages 18–35, the findings should be generalized only to populations reasonably similar to the participants studied.

Member Contributions

Christine Peng: Proposed the music and attention experiment that was selected as the group's primary study and collaborated with the group to develop the final study design and protocol. Participated in pilot testing and collected 15 usable observations during final data collection. Completed the data audit, documented deviations, created visualizations and descriptive summaries, reviewed the final dataset and reported results, and checked the analysis file. Christine was also responsible for the presentation timing check and final package submission.

Sam Reyes: Contributed to the initial experiment brainstorming and collaborated with the group to develop the final study design and protocol. Participated in pilot testing and collected 13 usable observations during final data collection. Contributed to the written report by completing the title, abstract, scientific question, and rationale. Sam was also responsible for designing and completing the presentation slides. Sam worked with Lucas to present the final presentation.

Lorena Rosales-Jaiver: Contributed to the initial experiment brainstorming and collaborated with the group to develop the final study design and protocol. Participated in pilot testing and collected 15 usable observations during final data collection. Contributed to the written report by completing the discussion, limitations, conclusion, and member contributions sections. Lorena was also responsible for integrating the final report and ensuring that all sections were complete and consistent with the overall study.

Lucas Tineo: Contributed to the initial experiment brainstorming and collaborated with the group to develop the final study design and protocol. Participated in pilot testing and collected 16 usable observations during final data collection. Contributed to the written report by completing the registered primary analysis and results sections. Lucas was also responsible for serving as the presenter for the final presentation along with Sam.

Wanxun Fu: Contributed to the initial experiment brainstorming and collaborated with the group to develop the final study design and protocol. Participated in pilot testing and collected 14 usable observations during final data collection.